

End-to-End HR Analytics Data Engineering Pipeline

1. Project Overview

- The HR Data Pipeline is an end-to-end batch data engineering project designed to simulate an HR analytics data platform.
- The system generates synthetic HR data using Python (Faker), processes daily employee snapshots, stores historical records in a relational warehouse and enables reporting via Power BI.
- The pipeline is fully containerized using Docker and orchestrated using Apache Airflow.

2. Objectives

- Design a scalable HR data warehouse using fact constellation schema modelling
- Implement automated ETL workflows using Airflow
- Maintain historical tracking of employee metrics
- Provide aggregated KPIs for BI reporting

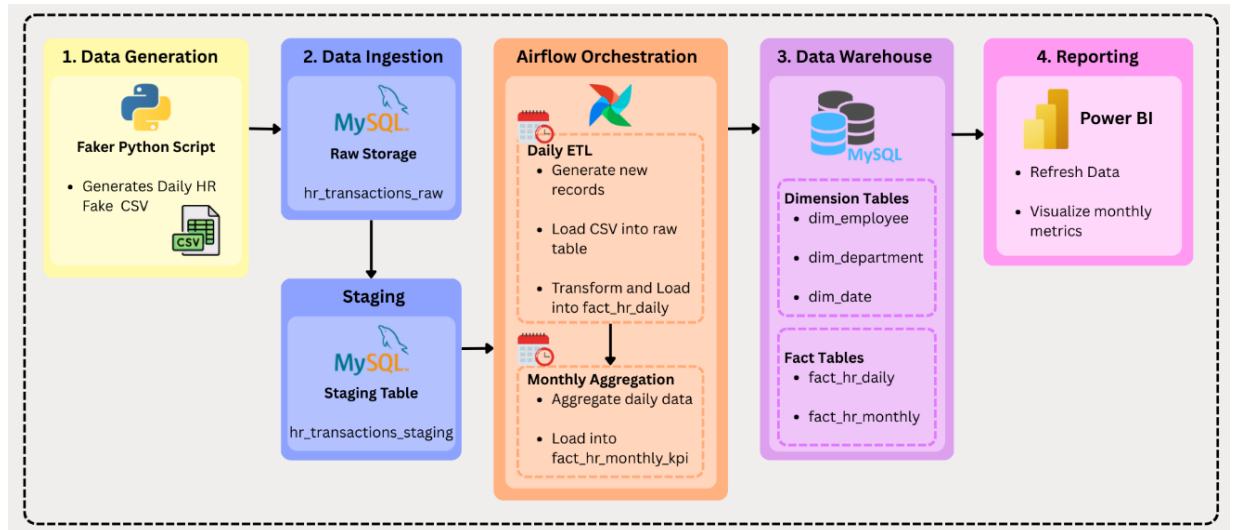
3. Technology Stack

Component	Technology
Data Generation	Python – Faker Library
Orchestration	Apache Airflow
Database	MySQL
Containerization	Docker + Docker Compose
BI Tool	Power BI
Development Environment	Local (Docker Desktop)

4. Data Volume

- Approximately 500 employees in master dataset
- Daily snapshot generation
- Historical accumulation over time

5. System Design



6. Data Flow Design

6.1 Raw Layer

Table: *hr_transactions_raw*

- Stores all historical daily snapshots
- Append-only
- No transformations applied
- This table acts as the raw landing zone for all daily HR data.

Table: *hr_transactions_staging*

- Stores daily snapshots
- Acts as a staging layer
- No transformations applied

6.2 Dimension Layer

Table: *dim_employee*

- EmployeeID (Primary Key)
- Gender
- DOB
- HireDate
- JobRole

Static employee attributes.

Table: *dim_department*

- DepartmentID (Auto Increment)
- DepartmentName (Unique)

Table: *dim_date*

- DateValue (Primary Key)
- Year
- Month
- Day
- MonthName
- Quarter

6.3 Fact Layer

Table: *fact_hr_daily*

Stores cleaned and transformed historical daily records.

Primary Key: EmployeeID, SnapshotDate

Contains:

- MonthlyIncome
- Overtime
- JobSatisfaction

- PerformanceRating
- Relieved
- Foreign keys to dimensions

6.4 Aggregate Layer

Table: *fact_hr_monthly*

Stores aggregated KPIs by:

- Year
- Month
- Department

Metrics:

- TotalEmployees
- AvgSalary
- AttritionCount
- Attrition Rate
- Overtime Rate
- AvgTenure

7. ETL Design

Daily Workflow

1. Generate daily snapshot CSV
2. Load data into *hr_transactions_raw* and *hr_transactions_staging*
3. Populate/update:
 - *dim_employee*
 - *dim_department*
 - *dim_date*

4. Insert cleaned records into *fact_hr_daily*

Monthly Workflow

1. Aggregate data from *fact_hr_daily*

2. Insert into *fact_hr_monthly*

3. Power BI refresh

8. Scheduling Strategy

- **Daily DAG:** Snapshot processing & fact loading
- **Monthly DAG:** KPI aggregation
- Managed by Apache Airflow Local execution via Docker containers